

Report with potential improvements Task 1

1. Data augmentation: Introducing hard negations (rivers, cities, and brand names containing the word "mountain") and using synthetic text generation to increase contextual diversity.
2. Architecture upgrades: Evaluating DeBERTa-v3 as a base model and integrating a CRF decoding layer to ensure correct transitions during sequence labeling.
3. Tokenization artifact handling: Expanding the tokenizer vocabulary with domain-specific entities to minimize subword splitting in non-English/transliterated mountain names.
4. Inference optimization: Exporting the model to ONNX format with 8-bit quantization to improve deployment efficiency and reduce latency in CPU-based environments.